

Spacetime Symmetry as Domain Anchor

Poincaré, Lorentz, Translation, and CPT

Kernel-Faithful, Constructor-Saturated, Semantically Causal,
Projection-Recompositional, Apex-Closed Normal-Form Closure

Central result. One fixed Minkowski/Wightman/state/sector/Noether/CPT target is classified. The target contains 128 typed primitive leaves and 18 executed bridge obligations. The occupant-free burden construction yields 13 canonical roles. A role-blind physical-object process builds 15 parent objects and reaches the fixed point of the authoritative constructor grammar, producing 39 physical facets, presentation variants, diagnostics, and target edges. Validated forward, inverse, and composition witnesses form 30 presentation residues. All 390 role–residue pairs are classified from primitive physical facts; 15 positive edges generate 2 complete pointwise-eligible preassignments, of which 1 is compatible. The assignment quotient under the induced live action of the full-certificate automorphism group has 1 orbit.

Every one of the 64 architectures generated by the registered refinement grammar is rebuilt from typed executable projections and verified recomposition maps and transports to the canonical orbit. Conservative certificate enrichments preserve that orbit; target-defining or standing-bearing enrichments reopen closure. The canonical and normalized presentations of the selected Jost/Wightman CPT theorem agree, while unbridged algebraic and Lagrangian formulations remain distinct certificate targets. The fixed local physical construction is non-degenerate and determinate, so kernel instantiation is necessary, complete, and independent of every downstream AMetric or apex consequence. The finite certificate determines one lawful local assignment orbit; the imported apex theorem then forces one admissible interior and excludes a second same-domain standing classifier or discrete plurality of admissible regimes.

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Abstract

The expression *spacetime symmetry* denotes several non-identical structures: the relativistic target, active transformations of states and observables, passive coordinate and frame redescription, projective state representations, finite-dimensional field covariance, particle-sector response, infinitesimal generators, Noether currents and charges, Ward identities, and the premise, theorem, and implementation loci of CPT. This article gives one fixed-target grammar for those loci and proves a stable Poincaré–CPT normal-form theorem.

AASC is not an optional interpretation of independently complete physics. The fixed local construction is non-degenerate and determinate: target, carrier, locus, comparison, transformation, and endpoint identities remain fixed; completed standing-positive and standing-negative constructions occur; and substantive bridge conditions exclude a complete rival assignment. Admissibility, Standing, Reference, and Irreversibility are therefore instantiated by necessity. Geometry and quantum field theory supply the target-specific propositions inside that already kernel-bearing domain; they do not create determinacy from a standing-neutral exterior.

The target separates Minkowski geometry and its presentations; positive-metric projective states; gauge-invariant Wightman fields and local operators; charged-sector controls; stable, resonant, confined, and composite sectors; self-adjoint Poincaré generators; an explicitly admitted Noether-current realization; boundary conditions for whole-space charges; and one canonical Jost/Wightman CPT packet. Roles and physical candidates are produced in isolated processes. The candidate universe is the fixed point of an authoritative constructor grammar, and every target-defining datum is classified as identity-fixing or causally used by construction, burden formation, compatibility, presentation, evidence, or target-update routing.

The executed certificate contains 128 target leaves, 27 burdens, 13 canonical roles, 15 parent objects, 30 presentation residues, and 390 classified role–residue pairs. 15 positive edges generate 2 complete pointwise-eligible preassignments; relational compatibility leaves 1 assignment and one lawful orbit. Full-certificate and live automorphism groups are computed separately. All 64 registered finite-interface refinements recover their parents and transport to the canonical orbit. All 18 completeness clauses and all 215 destructive, 15 guard/dependency, and 9 aggregate reconstruction checks pass.

Passive coordinate, frame, spin-frame, ray-phase, current-improvement, and CPT convention changes are skin only where executable witnesses preserve the complete physical tuple. Active translations, rotations, boosts, and CPT remain physical actions; mass, spin, helicity, causal structure, representation class, generators, and charges remain target-internal content. The certificate establishes the unique lawful local normal form of the fixed target. A proposition-exact downstream interface now includes the separately registered upstream AMetric-boundary necessity theorem and the apex compression in *Unus Solus Possibilis Est*; once the already-instantiated kernel, fixed domain, boundary package, and local singleton are fixed, the apex theorem forces one admissible interior and excludes a second same-domain standing classifier. No uniqueness claim is made for all relativistic quantum field theories.

Read This First: Dependency and Proof Spine

$$\text{NonDegDet}\left(D_{\text{SYM2}}^{\text{phys}}\right) \Longrightarrow K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}).$$

This implication is necessity, not a selectable interpretive premise. The finite certificate records the target-specific witnesses of non-degenerate determinacy; it does not generate the kernel.

Dependency lock: kernel activation is complete

Kernel activation is complete at the implication above. No AMetric result, apex theorem, source ledger, or software-closure condition is an additional condition for instantiating Admissibility, Standing, Reference, or Irreversibility. Everything below concerns target-specific physical certification or forced consequences of an already-instantiated kernel.

$$K_D + \text{fixed physical interfaces} \Longrightarrow D_{\text{SYM2}}^{\star} + C_{\text{phys}}.$$

$$D_{\text{SYM2}}^{\star} \Longrightarrow B_D \Longrightarrow R_D = B_D / \sim_{\text{gov}},$$

where the quotient is canonical relative to the frozen burden grammar. Independently,

$$D_{\text{SYM2}}^{\star} + C_{\text{phys}} \Longrightarrow O_D \xRightarrow{\mathcal{K}} F_D \Longrightarrow X_D = F_D / \sim_{\text{pres}}.$$

Here O_D is the role-blind parent-object registry, $\mathcal{K}$ is the authoritative physical-constructor grammar, F_D is its fixed-point facet universe, and X_D is the witnessed presentation residue.

$$(R_D, X_D, C) \Longrightarrow E_D, J_D, X_D^{\text{live}}, \Gamma_{\text{cert}}, \Gamma_{\text{live}}.$$

The restriction homomorphism

$$\rho : \Gamma_{\text{cert}} \longrightarrow \Gamma_{\text{live}}$$

induces the theorem-bearing action, and

$$|\Sigma_C(D) / \text{im } \rho| = 1.$$

Every architecture generated by the registered refinement grammar is rebuilt from typed finite-interface projections and verified recomposition maps and transports to the canonical orbit. Registered conservative enrichments preserve that orbit; new target or standing content produces ClosureReopened.

$$\text{NonDegDet}\left(D_{\text{SYM2}}^{\text{phys}}\right) \Longrightarrow K_D,$$

$K_D + \text{fixed-domain and AMetric corpus consequences} \Longrightarrow \text{complete apex application package},$
 $\text{complete apex application package} + \text{local singleton normal form}$
 $\Longrightarrow \text{one admissible interior by the imported apex theorem}.$

Kernel-first in necessity; QFT-first in record certification

The explanatory order begins with the necessity conditions of non-degenerate determinate objecthood. The evidentiary order for a physical record begins with ordinary geometry and QFT: target, transformation, quotient, spectrum, domain, current, boundary, and CPT premises are fixed before an AASC role is reported. This prevents role labels from manufacturing their premises without treating physical determinacy as kernel-free.

Publication endpoint

For the frozen target SYM-2 and a role-realization-complete certificate:

1. the physical construction is non-degenerate and determinate, so the AASC kernel is instantiated by necessity;
2. every target-defining primitive is typed as identity-fixing or causally consumed, and every bridge is resolved to an invoked compatibility function;
3. role and candidate construction are process-isolated, while the candidate universe is the fixed point of the authoritative registered physical-constructor grammar;
4. every registered refinement executes type-specific finite-interface projections and verified parent recomposition;
5. presentation equivalence is generated by validated forward, inverse, and composition witnesses, and eligibility and compatibility descend to witness orbits;
6. the assignment quotient under the induced live action of the full-certificate automorphism group is singleton;
7. after kernel activation is already complete, the downstream apex consequence interface and the faithful same-domain certificate license the imported terminal theorem, which forces exactly one admissible interior and excludes a second same-domain standing classifier.

Exact falsification routes

The result is reopened by a same-domain demonstration that the local physical construction is not determinate or is degenerate; a break in kernel necessity; a missing target-defining primitive; an unregistered applicable physical constructor; an omitted standing burden; an overbroad or incomplete presentation quotient; a false eligibility edge; a missing bridge or compatibility condition; an invalid projection or recomposition map; an unaccounted lawful automorphism; a second assignment orbit; or a failure of the same-domain application interface. Premise deletion, target change, or replacement of the fixed certificate is not a same-scope counterexample.

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Dependency Order, Necessary Kernel Activation, and Apex Scope

1.1 AASC is necessity, not an interpretive overlay

A spacetime-symmetry claim becomes available for determinate use only when its target, acting object, acted-upon object, transformation, comparison class, quotient, consequences, scope, and reportable endpoint remain fixed throughout the classification act. These are not optional descriptive conveniences. They instantiate

$$K_D = (\text{Admissibility, Standing, Reference, Irreversibility})$$

before any conventional spacetime or quantum-field-theoretic representative can carry theorem force. A physical domain containing determinate states, fields, carriers, records, transformations, or endpoints is therefore kernel-bearing by necessity [14].

Ordinary geometry and quantum field theory remain indispensable. They supply the Minkowski structure, group actions, state and field representations, spectral assumptions, self-adjoint generators, current and boundary propositions, and CPT theorem interfaces used by the certificate. Their role is faithful target-specific specialization inside a prior determinate domain; they do not create target identity or admissibility from outside it.

Blocked dependency inversion

The project does not first assume a kernel-free completed physical ontology and then attach an AASC vocabulary. It identifies the conditions under which a determinate physical target and theorem object can be fixed and used at all, and then certifies the Poincaré–CPT specialization within that standing-bearing regime.

1.2 The local physical construction is non-degenerate and determinate

Let $D_{\text{SYM2}}^{\text{phys}}$ denote the fixed physical Poincaré–CPT realization domain. Its certificate fixes a typed target, carrier and locus identity, admissible comparison class, transformation witnesses, eligibility and compatibility predicates, and a reportable assignment endpoint. It contains a standing-positive completed construction and a standing-negative completed construction: the canonical assignment satisfies every bridge, while the translation/Lorentz-swapped complete preassignment fails named compatibility conditions. The gate is therefore neither empty nor universal, and its exclusion is substantive rather than stipulative.

SYM2-T63 — Non-degenerate determinacy of the fixed physical domain

The local physical SYM-2 construction is determinate because target, carrier, locus, comparison, transformation, and endpoint identities are fixed throughout theorem-bearing use. It is non-degenerate because both admissible and inadmissible completed constructions occur, at least one lawful realization survives, at least one complete rival assignment is excluded by named relations,

and an excluded completed act cannot be repaired retrospectively as the same act.

Table 1.1: Non-degenerate determinacy and necessary kernel activation.

Application-regime property	Result	Role
fixed typed target	PASS	Determinate-domain witness
fixed carrier and locus identity	PASS	Determinate-domain witness
complete bivalent comparison	PASS	Determinate-domain witness
identity preserving redescription fixed	PASS	Determinate-domain witness
completed endpoint is reportable	PASS	Determinate-domain witness
standing positive construction exists	PASS	Non-degeneracy witness
standing negative construction exists	PASS	Non-degeneracy witness
gate is neither empty nor universal	PASS	Non-degeneracy witness
classification has substantive exclusion	PASS	Non-degeneracy witness
kernel instantiated by necessity	PASS	P-KERNEL-NECESSITY
activation complete before downstream apex package	PASS	Dependency lock
downstream apex or AMetric package required for kernel activation	NO	Downstream consequence only
theorem domain	D_phys_SYM2	finite faithful certificate of the physical domain, not an independent audit domain

SYM2-T64 — Necessary kernel activation

Since D_{SYM2}^{phys} is non-degenerate and determinate, Admissibility, Standing, Reference, and Irreversibility are instantiated by necessity. The finite certificate witnesses the local instantiation conditions; the upstream kernel theorem supplies the necessity implication. No additional framework-selection premise is introduced.

Kernel activation is complete at SYM2-T64

No AMetric theorem, apex theorem, source-ledger entry, or software-closure result is an additional condition for kernel instantiation. Those materials enter only downstream, as fixed-domain consequences, application-interface controls, or terminal uniqueness results within an already kernel-bearing domain.

Logical stage	Status	Exact role
Non-degenerate determinacy	Established locally	Supplies the sufficient antecedent for necessary kernel instantiation.
Kernel instantiation	Necessary and complete at SYM2-T64	Admissibility, Standing, Reference, and Irreversibility are already operative; no downstream premise activates them.
Apex unique-interior consequence	Downstream theorem application	Uses forced fixed-domain and AMetric consequences plus the local singleton normal form; it does not complete or repair kernel activation.

1.3 The certificate is not a second domain

The theorem domain is D_{SYM2}^{phys} . The machine-readable project is a finite faithful certificate C_{SYM2}^{cert} of that physical domain. Record names, row order, source identifiers, file paths, and implementation metadata have no authority to create or alter physical standing. The certificate proves target-specific occupation, compatibility, quotient, and normal-form facts; it does not replace physical standing with bookkeeping standing.

1.4 Cost of kernel denial

Denied kernel role	Consequence for SYM-2
Reference	No fixed Minkowski target, event, state, field, representation, transformation, generator, current, charge, or CPT implementation remains.
Standing	Formulas and operators remain notation without determinate downstream theorem use or reportability.
Admissibility	No boundary separates active action, passive redescription, theorem consequence, hidden selector, same-act repair, and target update.
Irreversibility	Later background, boundary, formulation, or representation information may rewrite a completed classification act retrospectively.

A critic retaining the same fixed target, comparison class, lawful transformations, and theorem-bearing consequences while denying these functions retains the kernel under another name. A critic removing them changes or destroys the theorem object.

1.5 Three sequential closure burdens

The proof preserves three logically distinct layers.

1. **Necessary kernel and gate layer.** Non-degenerate determinacy instantiates the kernel and fixed-domain standing regime by necessity.
2. **Local normal-form layer.** Target-derived roles and role-blind physical residues have one complete compatible assignment orbit.
3. **Apex interior layer.** Within the already standing-instantiated fixed physical regime, the apex theorem excludes a second same-domain admissible construction regime.

The boundary package is imported from *The AMetric Boundary: Necessity, Non-Parameterization, and Closure by Exhaustion* [15]; the third-layer uniqueness theorem is imported from *Unus Solus Possibilis Est* [16]. Its gate-uniqueness, no-repair, no-generator, no-independent-authorizer, target-fixation, and unique-admissible-interior results are not replaced by a finite list of rival classifiers. The finite checker verifies only the target-specific local normal form and the faithful same-domain application interface; it neither generates the kernel nor reconstructs the apex theorem from below.

1.6 Downstream apex consequence interface

Kernel activation is already complete before this section begins. The following interface fixes the downstream consequence chain from the already-instantiated kernel-bearing domain to the apex uniqueness theorem. It is not an activation package. The controlled chain is

$$\text{NonDegDet}(D) \implies K_D,$$

K_D + fixed-domain consequences + the upstream AMetric-boundary theorem $\implies$ the complete apex application package,
 complete apex application package + nonempty standing-positive side $\implies$ Theorems 6–7 of *Unus Solus Possibilis Est*.

The downstream provenance certificate registers the upstream AMetric-boundary necessity theorem as its own prerequisite node, distinct from the apex-layer compression in *Unus Solus Possibilis Est*, and records exact normalized statement, hypothesis, conclusion, locator, specialization, limitation, and prerequisite-edge digests for the complete consequence route. It recursively verifies the registered dependency graph and fixes exact bibliographic source identity. The bundled apex source is additionally SHA-256 verified; unbundled dependency sources are reported as bibliographically fixed, with their bytes explicitly not locally verified. Imported theorem truth remains external to the finite checker. The kernel-necessity row records provenance for the already-complete activation theorem; every remaining row is explicitly typed as downstream consequence, application interface, derivational support, or terminal apex result. This is dependency fixation, not a second activation condition or a second local derivation.

Table 1.2: Proposition-exact downstream apex provenance: normalized content, recursive prerequisite closure, bibliographic identity, and source-integrity level. Unbundled source bytes remain imported and are not locally hash-verified.

Dependency role	Logical stage	Proposition	Exact content	DAG	Source identity	Source integrity	Status
kernel necessity	KernelActivation	P-KERNEL-NECESSITY	PASS	PASS	PASS	Bibliographic; bytes unbundled	Verified
fixed domain no repair	DownstreamFixedDomainConsequence	P-FIXED-DOMAIN	PASS	PASS	PASS	Bibliographic; bytes unbundled	Verified
standing instantiated application anchor	ApexApplicationInterface	P-APEX-APPLICATION-ANCHOR	PASS	PASS	PASS	Local SHA-256 verified	Verified
intelligibility	ApexApplicationInterface	P-APEX-INTELLIGIBILITY	PASS	PASS	PASS	Local SHA-256 verified	Verified
irreversible commitment	ApexApplicationInterface	P-APEX-IRREVERSIBILITY	PASS	PASS	PASS	Local SHA-256 verified	Verified
admissible composition	UpstreamDerivationalSupport	P-APEX-COMPOSITION	PASS	PASS	PASS	Local SHA-256 verified	Verified
scope discipline	ApexApplicationInterface	P-APEX-SCOPE-DISCIPLINE	PASS	PASS	PASS	Local SHA-256 verified	Verified
meta invariance	ApexApplicationInterface	P-APEX-META-INVARIANCE	PASS	PASS	PASS	Local SHA-256 verified	Verified
ametric boundary necessity	DownstreamAMetricConsequence	P-AMETRIC-BOUNDARY-NECESSITY	PASS	PASS	PASS	Bibliographic; bytes unbundled	Verified
binary non eventful ametric boundary	DownstreamAMetricConsequence	P-APEX-AMETRIC-BOUNDARY	PASS	PASS	PASS	Local SHA-256 verified	Verified
bivalence	ApexApplicationInterface	P-APEX-BIVALENCE	PASS	PASS	PASS	Local SHA-256 verified	Verified
target fixation	ApexApplicationInterface	P-APEX-TARGET-FIXATION	PASS	PASS	PASS	Local SHA-256 verified	Verified
unique admissible interior	ApexTerminalConsequence	P-APEX-UNIQUE-INTERIOR	PASS	PASS	PASS	Local SHA-256 verified	Verified

Downstream apex consequence package

The apex conclusion is enabled only when every registered dependency role has exact normalized proposition content, exact prerequisite edges, a recursively closed path to the terminal theorem, exact bibliographic identity, and the declared source-integrity status. Any content, edge, source-metadata, or bundled-file corruption blocks apex status while leaving necessary kernel activation and the local normal-form theorem separately reportable.

1.7 Proof class and exact burden

This article is a kernel-forced, fixed-target, proposition-certified, finite normal-form theorem with an imported apex consequence. It does not derive special relativity, Minkowski signature, the Standard Model, interacting Wightman fields, renormalized currents, or the CPT theorem from empty premises. Those structures and propositions are fixed as explicit imported target interfaces.

An objection has theorem standing if it breaks kernel necessity while preserving the same non-degenerate determinate target, invalidates an imported proposition or its specialization, exposes an unconsumed target datum or bridge, identifies a missing burden or physical facet, breaks a presentation witness, eligibility edge, compatibility condition, refinement transport, automorphism action, or supplies a second faithful same-target assignment orbit. Vocabulary preference, premise deletion, or a changed target is not a same-scope counterexample.

Integrated Target, Semantic Use, and Fixed-Domain Discipline

2.1 The primary target

The theorem target is one integrated multi-interface object, SYM-2. It fixes simultaneously:

- four-dimensional oriented and time-oriented Minkowski geometry, causal order, vacuum, spectrum condition, and the connected Poincaré cover;
- coordinate, frame, and spin-frame presentations and passive redescription classes;
- a positive-metric projective state interface and strongly continuous connected action;
- a gauge-invariant Wightman-field target with local operator covariance and an explicitly registered BRST projection interface;
- charged-sector controls and distinct stable, massless, resonant, confined, and composite sector classes;
- a self-adjoint Poincaré generator system with a common invariant domain;
- an admitted renormalized local Noether-current package and a separate boundary/charge target;
- one canonical Jost/Wightman CPT premise packet and one physical implementation interface;
- neighboring curved, gravitational, Lorentz-violating, nonlocal, nonunitary, thermal, asymptotic, and quantum-gravity target edges.

Table 2.1: Integrated target interfaces.

Interface	Type	Fixed target content	Leaves
IF-GEOMETRY	geometry	Four-dimensional oriented and time-oriented Minkowski domain	9
IF-PRESENTATION	presentation	Coordinate, orthonormal-frame, and spin-frame presentation system	7
IF-PASSIVE	passive_ redescription	Passive coordinate, frame, and spin-frame redescription class	6
IF-POINCARÉ-ACTION	group_action	Connected Poincaré action on physical states and local fields	13
IF-STATE	state_ representation	Positive-metric projective state representation	9
IF-WIGHTMAN-FIELD	field_covariance	Gauge-invariant Wightman-field and local-operator covariance interface	14
IF-CHARGED-SECTORS	charged_sector_ control	Charged-sector conjugation control distinct from the canonical Wightman target	4
IF-SECTORS	sector_response	Stable, massless, resonant, confined, and composite sector partition	7
IF-GENERATORS	generator_bridge	Self-adjoint Poincaré generator system on a common invariant domain	9
IF-NOETHER-LOCAL	noether_local	Admitted renormalized local Poincaré-current and Ward-identity realization	8
IF-CHARGE-BOUNDARY	charge_ boundary	Whole-space charge and boundary/falloff realization	6
IF-CPT-PREMISE	cpt_premise	Canonical Jost/Wightman weak-local-commutativity premise architecture	13
IF-CPT-ACTION	cpt_action	Physical antiunitary CPT implementation on states and gauge-invariant local fields	7
IF-DIAGNOSTICS	diagnostics	Derived interval, Casimir, Ward, and CPT consequence witnesses	5
IF-TARGET-EDGES	target_edges	Registered neighboring spacetime and theorem targets	11

2.2 Target lock and semantic consumption

Target identity and physical use are distinct audit questions.

Identity-fixing use. A datum contributes to the immutable target and certificate digest. Changing it changes the theorem object or routes to a target update.

Constructive use. A datum changes a generated burden, parent object, facet, presentation witness, or physical relation.

Eligibility or compatibility use. A datum controls role incidence or cross-interface co-instantiation.

Evidence use. A datum discharges a hypothesis of an imported proposition.

Target-edge use. A datum explicitly routes a neighboring theory outside the fixed target.

Every primitive leaf is typed and read through a tracked accessor. No leaf is reported as physically consumed merely because it appears in a checksum. The destructive audit mutates every target-defining value and verifies a substantive certificate change or an explicit fail-closed disposition.

2.3 Typed quotient family

The target distinguishes at least:

$$Q_D^{\text{coord}}, \quad Q_D^{\text{frame}}, \quad Q_D^{\text{ray}}, \quad Q_D^{\text{field}}, \quad Q_D^{\text{current}}, \quad Q_D^{\text{CPTphase}}.$$

Coordinate and frame representatives may describe one geometric object; nonzero state vectors differing by a common phase may represent one ray; field bases and current improvements may be presentation-equivalent under exact target-relative witnesses. Active Poincaré- or CPT-related physical states are not identified merely because the theory is invariant.

Integrated-target fixation

Every SYM-2 claim is either a target-locus facet of SYM-2, a witnessed presentation of such a facet, a derived diagnostic, an incomplete certificate or countermodel, or an explicit target update. Changing background, field framework, boundary class, spectrum condition, CPT formulation, locality, unitarity, or sector assumptions cannot be used as same-act repair.

Independent Burden and Physical-Object Construction

3.1 Two isolated construction pipelines

The target-role and candidate universes are not coauthored inside one conclusion-bearing pass.

1. The **burden process** reads only typed target obligations, denial losses, continuation requirements, compatibility profiles, and failure routes. It has no candidate input.
2. The **physical-object process** reads only target interfaces, registered physical constructors, and imported physical propositions. It has no role or burden input.
3. The two outputs first meet in the downstream eligibility calculation.

Separate process manifests and digests record this independence. Mutating role rules leaves physical candidate formation unchanged; mutating the physical constructor grammar leaves the role output unchanged.

3.2 Parent objects, facets, and natural surplus

The physical process first constructs natural parent objects: Minkowski geometry, the connected Poincaré representation, the projective state realization, the field-covariance system, sector registry, generator system, current and charge packages, and CPT theorem and action packets. Target-locus projections then expose atomic facets.

Candidates retain all natural target-relevant physical facts and are not reduced to role-tailored minimal checklists. Twelve of the thirteen live occupants contain strict physical-fact surplus beyond the facts needed by their occupied roles. The presentation-system occupant is exact rather than surplus-bearing, but its formation remains role-blind and process-isolated. The complete surplus audit is machine-readable in the project.

3.3 Candidate kinds and upstream dispositions

The top-level candidate grammar remains

Action $\sqcup$ Structure $\sqcup$ Diagnostic $\sqcup$ Transport.

Each generated record also carries an upstream target relation:

- actual same-target facet;
- actual same-target rival;
- presentation variant;
- derived witness only;
- incomplete certificate or countermodel;
- target update.

Only actual same-target facets and rivals enter occupation competition. Presentation variants are quotiented; diagnostics remain consequences; countermodels test failure routes; target updates are routed before same-scope eligibility.

3.4 Three independent classification axes

Every record retains separate answers to:

1. which target role, if any, it occupies;
2. whether it is anchor, physical tensor/action, witness, skin, obstruction, or target update under ATS;
3. its SYM-0 quotient, quantum, boundary, vacuum, anomaly, and evidence status.

No axis is inferred merely from another. Exactness does not imply primitiveness; quotient triviality does not erase the represented physical architecture; theorem derivation does not make an implementation unreal.

Canonical Target-Burden Quotient

4.1 Human executive summary: thirteen standing functions

The symmetry architecture is easier to read when its thirteen canonical roles are grouped by what they do. Three roles fix the physical domain and distinguish it from its descriptions. Five roles realize the symmetry on states, fields, and physical sectors. Three roles connect actions to generators, local currents, and global charges. Two roles separate the premises of CPT from its physical implementation. The roles are not thirteen names for one object; they are thirteen distinct standing functions that must be occupied coherently.

Table 4.1: Human map of the thirteen canonical standing roles.

Block	Human role	Standing function
Domain and description	Geometric domain	Fixes the Minkowski/Poincare spacetime target, causal order, orientation, and spin cover.
Domain and description	Presentation system	Supplies the coordinate, frame, and spin-frame atlases used to display the same target.
Domain and description	Passive redescription	Reexpresses coordinates or frames while preserving event and field identity.
Physical realization	Translation action	Implements physical spacetime translations on the positive-metric state space.
Physical realization	Lorentz action	Implements physical rotations and boosts on the same state representation.
Physical realization	Projective state realization	Carries the unitary or antiunitary projective action on physical rays.
Physical realization	Local field covariance	Transforms local field components and arguments without being identified with the ray action.
Physical realization	Sector response	Classifies massive, massless, resonant, confined, and composite sectors in the certified state space.
Generators and consequences	Poincare generator bridge	Links the active translation and Lorentz actions to their self-adjoint infinitesimal generators.
Generators and consequences	Noether local architecture	Supplies local currents, Ward identities, and admitted improvement freedom.
Generators and consequences	Global charge and boundary realization	Adds integrability, whole-space falloff, and vanishing boundary flux needed for global charges.
CPT	CPT theorem premise	Fixes the Jost/Wightman premise packet that licenses the CPT theorem.
CPT	CPT physical implementation	Implements the antiunitary CPT action on the certified physical states and local fields.

Geometry fixes the target; presentation systems display it; passive maps redescribe it without acting physically. Translation and Lorentz actions act on states, while projective state realization, local field covariance, and sector response occupy different representational loci. The generator bridge connects finite transformations to infinitesimal structure. Noether currents and global charges remain distinct because global standing additionally requires integrability and boundary control. CPT premises remain distinct from the antiunitary action they license.

The full content-addressed role identifiers, burden members, primitive-fact criteria, governance profiles, and partition records appear in the Technical Audit Supplement, Appendix E, and in the machine-readable project

archive.

4.2 Standing burdens

A standing burden is forced when a fixed target datum performs a standing-bearing function whose denial destroys the declared realization and cannot be reduced to a presentation change or downstream witness. Each burden records:

- the target paths that trigger it;
- the facts a faithful occupant must realize;
- the denial loss;
- continuation, compatibility, and failure profiles;
- its target locus.

No burden record contains an expected occupant or candidate identity.

4.3 Governance equivalence and canonical quotient

For burdens b, b' , governance equivalence is equality of the fixed denial, compatibility, continuation, failure, and locus-family profile:

$$b \sim_{\text{gov}} b'.$$

The canonical role architecture is

$$R_D = B_D / \sim_{\text{gov}}.$$

This quotient is canonical once the frozen target-burden grammar is fixed. The project does not claim that no future same-target burden refinement could ever be discovered; a nonisomorphic complete refinement is an explicit falsifier that reopens closure.

The complete canonical-role table, granularity audit, and role-architecture uniqueness certificate are supplied in the Technical Audit Supplement, Appendix E.

4.4 Granularity and refinement discipline

Temporal and spatial translation, rotation and boost, massive and massless sectors, stress-energy and angular-momentum currents, generator components, and state/field CPT implementation remain distinct physical facets. Whether they form separate canonical roles or a lawful refinement is decided by their denial, continuation, and compatibility profiles rather than by vocabulary.

The registered split, merge, omission, duplication, and target-update constructors are fully dispositioned in the supplement. Their executed result is summarized by the theorem below.

Canonical burden quotient

Within the frozen burden grammar, the standing-role architecture is the unique governance quotient up to reindexing. Every registered split, merge, omission, duplication, and target-update constructor is explicitly dispositioned. A complete nonisomorphic same-target partition remains a theorem-breaking refinement, not a case to be silently absorbed.

Metricity, Causal Structure, and Licensed Spacetime Realism

5.1 A Metric no-selection

A metric cannot authorize its own standing, choose its target, or create the reference identity required for its use. This is not an argument against metric reality. Once licensed, metricity faithfully realizes interval, causal, orientation, and comparison structure [12, 13].

5.2 Minkowski projection anchor

The target fixes

$$(M, \eta, o, t_o), \quad \eta = \text{diag}(1, -1, -1, -1),$$

together with causal order, orientation, time orientation, connected isometry structure, spin cover, invariant vacuum, and forward-spectrum interface. The coordinate matrix is a presentation; the invariant geometric packet is the domain anchor.

5.3 Coordinates, frames, and real content

Coordinates, orthonormal frames, and spin frames are required presentations. Their differences are skin only when a validated witness preserves geometry, event identity, state, field/operator content, continuation, and failure profile. The presentation architecture remains load-bearing because local formulas must assemble into one target.

Object	Type	Reason
Minkowski metric and causal packet	Anchor; physical tensor	Changing it changes target identity and invariant relations.
Metric components in one chart	Presentation skin	Components change under basis transformation while the tensor remains fixed.
Coordinate/frame/spin-frame atlas	Carrier constraint	Required for coherent local comparison; no chart has privileged standing.
Passive frame change	Quotient-trivial skin	Same geometry, event, state, and observable are represented differently.
Active Lorentz action	Physical action	The transformed state or observable is not removed by the presentation quotient.

SYM2-T06 — Licensed metric realism

A witnessed presentation transformation may change coordinates, frames, or components while preserving the real geometric packet. A variation changing interval, causal order, orientation, state, operator, representation, boundary, continuation, or failure data changes physical or target content and is not skin.

Active and Passive Poincaré Transformation Noncollapse

6.1 Four objects under one name

A “Lorentz transformation” may denote target isometry structure, an active state or operator action, a passive coordinate/frame redescription, or finite-dimensional field-component covariance. Their formulas are related; their loci and quotients are not.

6.2 Passive redescription

For a coordinate or frame change representing the same geometric event and field,

$$q_D^{\text{pres}}(g \cdot r) = q_D^{\text{pres}}(r).$$

The operation is skin only when an explicit witness preserves the complete target and physical tuple.

SYM2-T07 — Passive presentation theorem

An invertible coordinate, frame, or spin-frame change is quotient-trivial exactly when a validated presentation witness preserves geometry, event identity, physical ray, local field content, boundary status, continuation, and failure data.

6.3 Active physical action

A connected Poincaré action maps physical rays and local fields,

$$[\psi] \mapsto [U(a, \Lambda)\psi], \quad U(a, \Lambda)O_i(x)U(a, \Lambda)^{-1} = D_i^j(\Lambda^{-1})O_j(\Lambda x + a),$$

with convention-dependent placement of inverses. The transformed physical state is not generally identified with the original by coordinate, basis, or phase equivalence.

SYM2-T08 — Active/passive noncollapse

At fixed target, active translations, rotations, and boosts are quotient-nontrivial physical facets. Passive coordinate and frame transformations are witnessed presentation facets. Neither status follows from matrix notation alone.

Claim status. Imported representation and QFT propositions plus local quotient classification [1, 23, 24].

6.4 Fixed points do not trivialize an action

An invariant vacuum is a fixed point, not evidence that the group acts trivially on every state. Stabilizers and invariant subspaces coexist with nontrivial transformations.

General covariance is not Poincaré action

Passive coordinate covariance, active Poincaré invariance, diffeomorphism gauge symmetry, and local Lorentz gauge symmetry belong to different targets and quotient structures.

Projective State Symmetry and Unitary/Antiunitary Implementation

7.1 Physical rays

Pure states are rays rather than nonzero Hilbert vectors:

$$\mathcal{Q}_D^{\text{ray}} = (\mathcal{H} \setminus \{0\})/\mathbb{C}^\times.$$

A common vector phase is skin at this locus; transition probabilities and relative interference data remain physical.

7.2 Wigner implementation

A bijection of rays preserving transition probabilities admits a linear unitary or antilinear antiunitary implementation [20]. Bargmann's projective-representation analysis then controls continuous group multipliers [1].

SYM2-T09 — Ray-implementation bridge

Transition-probability-preserving ray symmetry supplies a unitary or antiunitary Hilbert-space implementation. Strong continuity through the identity selects the unitary branch for the connected Poincaré component; the antiunitary branch remains available for discrete operations such as the selected CPT implementation.

7.3 Generator domains

Strong continuity is not enough by itself to license formal generator algebra. The target includes self-adjoint generators on a common invariant domain, and compatibility requires exponentiation to the selected action [21].

Phase is not physical action

Multiplication of one vector by a common phase does not change its ray. A phase convention can be physically relevant only through a separate sector, interference, or operator certificate.

Local Field Covariance and Representation Loci

8.1 State and field representations

The physical Hilbert-space representation and the finite-dimensional field covariance matrices occupy different loci:

$$U(a, \Lambda) \neq D(\Lambda).$$

The former acts unitarily on physical states in the connected component; the latter transforms field components and need not be unitary in that sense [23].

8.2 Canonical field target

The canonical local-field interface consists of gauge-invariant Wightman fields or local observable-valued distributions on a positive-metric Hilbert space, with Lorentz covariance, adjoint closure, temperedness, invariant vacuum, forward spectrum, and weak local commutativity. Charged superselection sectors and gauge-fixed/BRST fields are separately typed controls, not silently included in the canonical CPT packet.

8.3 Argument and component transformations

Local covariance separates the transformed spacetime argument from the finite-dimensional component action. Similarity transformations of gamma matrices and spinor bases are presentation skin only when the simultaneous field and Clifford relations are preserved by a validated witness.

SYM2-T10 — State/field representation noncollapse

Projective state representation, Hilbert-space unitary implementation, finite-dimensional field covariance, gamma/spinor-basis choice, and local observable content are distinct. Equivalent bases are witnessed presentations; the representation response they display is physical.

Translation Action and Energy–Momentum Architecture

9.1 Translation facet

The connected Poincaré parent action has the canonical subgroup projection

$$\pi_{\text{trans}} : U(a, \Lambda) \mapsto U(a, I).$$

This projected facet acts physically on rays and local fields. The parent object remains a support composite rather than being duplicated as two whole occupants.

SYM2-T11 — Translation physical-action theorem

The target-derived translation facet is quotient-nontrivial on physical rays and local fields. Passive shifts of the coordinate origin belong to the presentation groupoid and do not occupy the translation-action role.

9.2 Energy–momentum generators

Under the declared continuity and common-domain assumptions,

$$U(a) = \exp(-ia_\mu P^\mu).$$

The finite action, self-adjoint generator, local stress-energy current, integrated charge, and Ward identity remain distinct records. Their bridges are checked relationally.

9.3 Spectrum and vacuum

The target fixes an invariant vacuum and forward spectrum. These conditions constrain the representation without making the translation action quotient-trivial.

Rotations, Boosts, and the Poincaré Generator Bridge

10.1 Lorentz facet

The parent Poincaré action induces

$$\pi_{\text{Lor}}: U(a, \Lambda) \mapsto U(0, \Lambda),$$

with rotation and boost subfacets. They share one Lorentz-action governance role because, at the declared target level, their denial profiles and compatibility obligations are the two components of one connected Lorentz response. A distinct standing function would reopen the granularity theorem.

SYM2-T12 — Rotation/boost action theorem

Rotations and boosts are quotient-nontrivial physical facets of one connected Lorentz action. Their rapidity, axis, and component coordinates are presentation data; their action on physical rays and fields is not.

10.2 Generator algebra

The generator bridge contains P^μ and $M^{\mu\nu}$ on a common invariant domain and satisfies the translation, Lorentz, and mixed Poincaré commutators. Compatibility verifies that these generators exponentiate to the selected subgroup actions.

Table 10.1: Poincare generator bridge.

Generator facet	System	Action IDs	Domain/algebra facts
Poincare generator bridge	GEN-POINCARE-1	ACT-TRANS-1; ACT-LOR-1	common_invariant_domain; cross_poincare_commutators; exponentiates_to_actions; lorentz_commutators; lorentz_generators; self_adjoint_generators; strongly_continuous_action; translation_commutators; +2 more

SYM2-T13 — Action/generator noncollapse

Finite group action, infinitesimal generator, common domain, commutator algebra, and exponentiation relation are independent theorem objects. A formal operator symbol cannot occupy the generator role without the declared domain and bridge certificates.

Particle and Sector Response: Mass, Spin, Helicity, Resonance, and Confinement

11.1 Sector partition

The target separates:

1. stable massive asymptotic sectors;
2. stable massless helicity sectors;
3. unstable resonance sectors;
4. confined field representations;
5. composite asymptotic sectors.

Only the first two receive strict Wigner one-particle classification [23, 24].

11.2 Mass and Pauli–Lubanski invariants

For stable sectors, the Poincaré Casimirs classify mass and spin or helicity. These values are physical representation response, not coordinate labels. Resonance pole data, confined fields, and composites are separately typed rather than promoted to stable irreducible one-particle representations.

Table 11.1: State, field, and sector representation noncollapse.

Representation locus	Selected facet	Locus	Certified physical facts
Projective State Realization	Projective state realization phase π	Representation/State Space	acts_on_physical_rays; antiunitary_discrete_branch; forward_spectrum_declared; positive_metric; projective_representation; ray_quotient_defined; strongly_continuous_action; +3 more
Local Field Covariance	Local field covariance gamma basis A	Representation/LocalFields	acts_on_local_fields; adjoint_closure; argument_transform; brst_projection_registered; component_transform; local_operator_sector_registered; lorentz_covariant_fields; +8 more

continued

Representation locus	Selected facet	Locus	Certified physical facts
Asymptotic Sector Response	Asymptotic and nonasymptotic sector response	Representation/ SectorSpace	charged_sector_ framework_registered; composite_sector_ typed; confined_field_ typed; conjugate_ sectors_registered; finite_statistics_ conditional; sector_ distinctions_preserved; stable_massive_sector; +2 more

SYM2-T14 — Sector-response theorem

Mass, spin, and helicity are quotient-stable physical response in the stable asymptotic sectors admitted by the target. Unstable resonances, confined fields, and composite asymptotic states remain distinct. Field covariance, stable-particle representation, resonance poles, and confinement status are not inferred from one another.

Noether Currents, Ward Identities, Charges, and Boundaries

12.1 Six non-identical objects

The target distinguishes

$$\text{action} \neq \text{generator} \neq \text{current} \neq \text{charge} \neq \text{Ward identity} \neq \text{improvement}.$$

Their relations are theorem- and boundary-dependent.

12.2 Explicit target condition

The manuscript does not infer a renormalized local stress-energy tensor from abstract Poincaré covariance alone. The selected target explicitly admits a renormalized local Poincaré-current and Ward-identity realization, together with whole-space falloff, integrable charges, zero flux, and charge-generation relations. SYM-2 classifies this admitted package; it does not construct it from first principles [10, 18].

12.3 Improvement witnesses and boundary routing

Canonical, Belinfante, and improved representatives may encode the same local and global content under suitable conditions [2, 3]. An improvement is presentation-equivalent only when its witness preserves local identities, renormalized operator content, integrated charge, boundary flux, generator relation, continuation, and failure profile. Regge–Teitelboim is used only for differentiability and surface-term discipline, not as a universal flat-QFT current theorem [19].

Table 12.1: Noether local/global noncollapse.

Layer	Selected facet	Certified content	Scope qualification
Local current/ Ward	Renormalized Noether current package (canonical)	admitted_current_package; angular_momentum_current; charges_generate_actions; current_ existence_explicit_target_data; improvement_family; integrable_ charges; local_divergence_relations; stress_energy_current; +1 more	Explicit admitted current target
Global charge/ boundary	Whole-space charge and boundary realization	admitted_current_package; charges_generate_actions; finite_ region_projection_registered; integrable_charges; ward_identities; whole_space_falloff; zero_flux_at_ infinity	Whole-space falloff target

Table 12.2: Boundary-relative Noether normal form.

Noether normal-form property	Result
Current existence status	Imported target interface
Boundary-relative improvement class	Witnessed
Canonical assignment orbit	1

SYM2-T15 — Noether local/global noncollapse

For the target containing the declared current realization, local current, Ward identity, integrated charge, boundary flux, and generation of the physical action are independent coordinates. A local identity may hold while a global charge is undefined, nonintegrable, or nonconserved because of flux.

SYM2-T16 — Boundary-relative improvement quotient

Two current representatives belong to one presentation class exactly when a validated improvement witness preserves the complete local and boundary-relative consequence tuple.

SYM2-T22 — Boundary-relative Noether normal form

For the fixed admitted current and boundary target, action, generator, renormalized current, integrated charge, Ward identity, and validated improvement representatives have one compatible locus-sensitive normal form. Changing boundary conditions or charge integrability is a target change or a distinct physical charge realization, not a presentation repair.

CPT Premise Architecture, Theorem Consequence, and Physical Action

13.1 Canonical theorem packet

The primary theorem packet is CPT-JOST-WLC, a Jost/Wightman weak-local-commutativity formulation over the positive-metric gauge-invariant Wightman-field target. Its premises are separately registered: Minkowski geometry and orientation data, Lorentz covariance, positive metric, invariant vacuum, forward spectrum, tempered field distributions, adjoint closure, weak local commutativity at Jost points, the required analytic interface, and the separately typed spin–statistics dependency [9, 11, 22].

SYM2-T17 — CPT premise-completeness theorem

A CPT theorem record is complete only when one exact formulation, target framework, premise set, conclusion, source proposition, and failure route are fixed. Deleting a premise produces an incomplete certificate; substituting another theorem framework produces a distinct certificate target unless a proposition-level bridge is established.

13.2 Theorem implication and physical implementation

The theorem implication is a diagnostic consequence about the selected class of theories. The physical CPT facet is an antiunitary implementation acting on physical states and gauge-invariant local fields while preserving the dynamics. These are non-identical records.

SYM2-T18 — CPT theorem/action separation

The premise architecture, theorem implication, antiunitary physical implementation, charged-sector conjugation consequence, and implementation-phase convention occupy different loci. The theorem statement is not the action; the action is not presentation skin; the phase convention is not a second physical operation.

13.3 Charged-sector control

Charge-conjugate sector mapping is treated through a separate conditional bridge with superselection and finite-statistics hypotheses [7, 8]. It is not silently inserted into the canonical local-field premise packet.

13.4 CPT relation lattice

The formulation registry distinguishes six relation types: same presentation orbit, theorem equivalence, consequence-preserving transport, shared conclusion with distinct premises, distinct certificate target, and physical target update. One certified bridge relates two presentations of the same Jost/Wightman theorem. Algebraic modular PCT and the Greaves–Thomas Lagrangian theorem remain distinct certificate targets [5, 17].

Table 13.1: CPT formulation registry.

Formulation	Name	Status	Target	Family	Premises	Conclusion
CPT-JOST-WLC	Canonical Jost/Wightman weak-local-commutativity formulation	Canonical	GaugeInvariantWightmanFieldTarget	JOST-WLC	minkowski; lorentz_covariance; weak_local_commutativity; positive_metric; unique_vacuum; +5 more	CPT-EXISTENCE-1
CPT-JOST-WLC-NORMALIZED	Normalized presentation of the canonical Jost/Wightman theorem	CertifiedEquivalent Presentation	GaugeInvariantWightmanFieldTarget	JOST-WLC	minkowski; lorentz_covariance; weak_local_commutativity; positive_metric; unique_vacuum; +5 more	CPT-EXISTENCE-1
CPT-ALGEBRAIC-MASSIVE	Algebraic modular massive-theory PCT formulation	DistinctCertificateTarget	AlgebraicMassiveNetTarget	ALGEBRAIC-MODULAR	haag_kastler_net; massive_sector; modular_covariance; finite_statistics_conditions	ALGEBRAIC-PCT-CONTROL
CPT-GT-LAGRANGIAN	Greaves-Thomas rigorous Lagrangian CPT formulation	DistinctCertificateTarget	LagrangianTensorTarget	LAGRANGIAN-CPT	complex_lorentz_covariance; formal_lagrangian_conditions	LAGRANGIAN-CPT-CONTROL

Table 13.2: Certified CPT presentation covariance.

Bridge	Source	Target	Premise bijection	Conclusion	Same theorem	Status
CPT-BRIDGE-JOST-NORMALIZATION	CPT-JOST-WLC	CPT-JOST-WLC-NORMALIZED	PASS	PASS	PASS	Certified

Table 13.3: CPT relation lattice and noncollapse.

Left formulation	Right formulation	Certified relation
CPT-JOST-WLC	CPT-JOST-WLC-NORMALIZED	SamePresentationOrbit
CPT-JOST-WLC	CPT-ALGEBRAIC-MASSIVE	DistinctCertificateTarget
CPT-JOST-WLC	CPT-GT-LAGRANGIAN	DistinctCertificateTarget
CPT-JOST-WLC-NORMALIZED	CPT-ALGEBRAIC-MASSIVE	DistinctCertificateTarget
CPT-JOST-WLC-NORMALIZED	CPT-GT-LAGRANGIAN	DistinctCertificateTarget
CPT-ALGEBRAIC-MASSIVE	CPT-GT-LAGRANGIAN	DistinctCertificateTarget

SYM2-T19 — Certified CPT presentation covariance

The canonical and normalized presentations of the same Jost/Wightman theorem induce the same lawful normal form. No equivalence with the algebraic modular or Lagrangian formulations is asserted.

SYM2-T20 — Conditional formulation-covariance rule

For any two CPT formulations, a proposition-level equivalence bridge preserving target, premises, conclusion, implementation class, continuation, and failure profile induces lawfully equivalent normal forms. Without such a bridge, the formulations remain distinct certificate targets.

SYM2-T23 — CPT relation-lattice closure

Presentation equivalence, theorem equivalence, consequence-preserving transport, shared conclusions under distinct premises, distinct certificate targets, and physical target updates are not collapsed. Every registered formulation pair receives one relation-lattice disposition, and concrete normal-form covariance is asserted only at the strength actually certified.

Target Updates and Neighboring Spacetime Regimes

14.1 Target-difference rule

An alternative is routed before same-target occupation whenever it changes geometry, field or state framework, positivity, locality, spectrum, sector category, boundary conditions, or CPT formulation.

SYM2-T21 — Spacetime target-update theorem

Curved background, dynamical gravity, diffeomorphism gauge symmetry, local Lorentz gauge in a tetrad target, Lorentz- or CPT-violating EFT, nonlocality, nonunitarity, thermal/media structure, BMS/asymptotic symmetry, quantum-gravity structure, or an unbridged CPT theorem formulation changes at least one target or certificate coordinate.

Lorentz-violating Standard-Model extensions are explicit neighboring EFTs [4]. Greenberg's result is used only in its one-way interacting-theory scope: CPT violation implies Lorentz violation under its assumptions; the converse is not asserted [6].

Table 14.1: Registered target updates.

Neighboring target	Target difference	Continuation	Disposition
Target edge: fixed_curved_background	edges.fixed_curved_background	TargetUpdate	Target Difference
Target edge: dynamical_gravity	edges.dynamical_gravity	TargetUpdate	Target Difference
Target edge: diffeomorphism_gauge	edges.diffeomorphism_gauge	TargetUpdate	Target Difference
Target edge: local_lorentz_gauge	edges.local_lorentz_gauge	TargetUpdate	Target Difference
Target edge: lorentz_violating_eft	edges.lorentz_violating_eft	TargetUpdate	Target Difference
Target edge: cpt_violating_eft	edges.cpt_violating_eft	TargetUpdate	Target Difference
Target edge: nonlocal_qft	edges.nonlocal_qft	TargetUpdate	Target Difference
Target edge: nonunitary_qft	edges.nonunitary_qft	TargetUpdate	Target Difference
Target edge: thermal_medium	edges.thermal_medium	TargetUpdate	Target Difference
Target edge: bms_asymptotic	edges.bms_asymptotic	TargetUpdate	Target Difference
Target edge: quantum_gravity	edges.quantum_gravity	TargetUpdate	Target Difference

No same-act repair

Adding a preferred vector, new boundary degree of freedom, altered positivity condition, nonlocal interaction, curved metric, or different theorem formulation after classification creates a distinct target or certificate state. It cannot retrospectively repair the completed act.

Physical Construction and Facet Residue

15.1 Typed target use

Every primitive target datum has a stable path, a controlled type, an allowed-value specification, and one or more declared uses. The audit distinguishes target-identity fixation from causal use in burden construction, physical construction, eligibility, compatibility, presentation, evidence matching, and target-edge routing. A read or checksum contribution alone is not reported as physical consumption.

Supplementary target-use registry

The complete target-leaf dependency ledger is printed in the Technical Audit Supplement, Appendix K. The article reports only the theorem-bearing totals and conclusions.

SYM2-T66 — Semantic target-use completeness

Every target-defining datum is either identity-fixing or causally connected to a theorem-bearing construction, burden, incidence decision, compatibility condition, witness, imported-proposition match, or target-edge disposition. Value mutations that change the declared target cannot leave the same completed certificate silently in place.

15.2 Executed bridge obligations

Every bridge obligation names a registered compatibility function. For every complete pointwise-eligible preassignment, the checker records the selected source and target facets, the hypotheses examined, the result, and the failure route. An unknown, missing, uninvoked, or false bridge prevents closure.

Supplementary bridge registry

All executed bridge obligations, inputs, dispositions, and failure routes are printed in the Technical Audit Supplement, Appendix K.

15.3 Role-blind parent-object and candidate formation

Role construction and physical construction run in isolated processes. The role process receives target burdens and denial profiles but no candidates. The physical process receives target interfaces, physical parent declarations, imported physical facts, and the constructor grammar but no roles, required-fact predicates, or intended assignment.

The physical universe begins with 15 admitted parent objects. Registered constructors are applied to all type-compatible inputs until a fixed point is reached. Each constructor application yields an emitted record, a certified duplicate, an inapplicability certificate, an incomplete-premise disposition, or a target update. Empty or incomplete required grammars fail before eligibility is evaluated.

SYM2-T65 and T67 — Constructor saturation and role-blind formation

The candidate universe is the finite fixed point of the authoritative registered physical-constructor grammar. Every generated object or facet has a constructor derivation, and no undeclared manual emission path is accepted. Candidate output is invariant under role renaming, role reordering, and alteration of downstream eligibility predicates.

15.4 Composite objects and atomic target-locus facets

A composite physical object may support more than one standing-bearing function. The connected Poincaré representation, for example, supports translation, Lorentz, state, field, generator, and sector facets without being duplicated as several whole objects. The support relation

$$c \triangleright f$$

records the parent, projection type, target locus, inherited physical content, continuation data, and failure profile of each facet.

Supplementary projection registry

The typed projection domains, codomains, preservation conditions, and parent-recovery records are printed in the Technical Audit Supplement, Appendix K.

15.5 Executable typed projection and recomposition

The registered refinement grammar uses typed maps rather than symbolic labels. Group restrictions, translation decompositions, Lorentz-chart refinements, generator components, representation-sector projections, Noether package decompositions, and CPT dependency decompositions each have declared domains, codomains, preservation requirements, executable projections, recomposition maps, and parent-recovery checks.

For every one of the 64 registered architectures the project executes the projections, generates the refined facets, rebuilds presentation classes and roles where required, recomputes eligibility, compatibility, assignments, automorphisms, and orbits, and verifies transport back to the canonical normal form.

Supplementary refinement audit

The complete 64-architecture reconstruction and orbit-transport ledger is printed in the Technical Audit Supplement, Appendix X.

SYM2-T69–T70 — Projection–recomposition adequacy and refinement stability

Every registered refinement is realized by validated type-specific projection and recomposition maps. Parent recovery, target preservation, bridge coherence, and orbit transport are checked. Every registered refined architecture has one lawful normal-form orbit canonically bijective with the unrefined orbit.

15.6 Generated universe and exact scope

Supplementary constructor trace

The generated-universe coverage table and full constructor trace are printed in the Technical Audit Supplement, Appendix K.

Completeness is relative to the frozen target, parent-object declarations, and registered constructor grammar. A new applicable constructor, parent object, target datum, or standing-bearing physical relation changes the certificate and reopens closure; it is not absorbed into the completed result.

15.7 Witnessed presentation groupoid

Presentation equivalence is generated by executable coordinate/frame, passive-coordinate, projective-phase, spinor-similarity, current-improvement, CPT-normalization, and CPT-phase witnesses acting on the actual candidate records. Forward and inverse maps, finite compositions, target preservation, and classifier covariance are checked.

Supplementary presentation certificate

The complete witness-groupoid, inverse, composition, and classifier-covariance ledger is printed in the Technical Audit Supplement, Appendix Y.

SYM2-T68 — Integrated witness-groupoid closure

The presentation quotient is the orbit relation generated by the validated finite witness groupoid on the frozen facet universe. Eligibility and compatibility descend to the quotient. No equivalence is inferred merely from identical names, source records, or downstream classification labels.

Eligibility, Relational Compatibility, and Dual Automorphisms

16.1 Complete eligibility incidence

Only after the target-derived role universe and role-blind presentation residue are independently frozen is every role–residue pair evaluated. Eligibility depends on primitive physical facts, target locus, and upstream candidate relation; it does not depend on display name, source identity, row order, or intended assignment.

Supplementary eligibility registry

The complete incidence summary and role–residue matrix are printed in the Technical Audit Supplement, Appendix M.

16.2 Registered physical alternatives and controlled failures

Presentation variants are removed by the witnessed quotient. The authoritative constructor grammar generates nine alternative records, all of which are witnessed presentation variants; it generates no genuine nonpresentation same-target rival residue. Countermodels, incomplete certificates, and target updates remain separately typed and cannot inflate the same-target alternative envelope.

Supplementary alternative envelope

The registered presentation alternatives, countermodels, and target-update dispositions are printed in the Technical Audit Supplement, Appendix Y.

SYM2-T73 — Registered same-target alternative classification

Every same-target alternative generated by the registered physical grammar is classified before assignment. In the present grammar, all nine alternatives are presentation variants and collapse through validated witnesses; the genuine nonpresentation rival set is empty. Controlled countermodels and neighboring targets are not reported as physical rivals.

16.3 Relational compatibility

Pointwise eligibility asks whether an individual residue can satisfy an individual role. Compatibility asks whether the selected occupants can jointly describe one physical realization. The certificate executes exactly 18 named bridge obligations, organized in Table 16.1 into nine reader-facing families.

Table 16.1: Reader-facing organization of the eighteen executed bridge obligations.

Reader-facing family	Executed obligations	Decisive condition	What it prevents
Geometry and presentation identity	2 of 18	The presentation atlas and active action refer to the same geometry, orientation, spin cover, and target.	An atlas or action for a different spacetime cannot enter the same certificate.
Active/passive separation	1 of 18	Passive maps preserve event and field identity and carry no nontrivial physical state action.	A coordinate or frame relabelling cannot occupy an active symmetry role.
State and field realization	3 of 18	The projective state action and finite-dimensional field covariance share one Poincare realization without being identified.	Projective Hilbert-space action cannot be replaced by a field-component representation, or conversely.
Sector synchronization	1 of 18	The physical state space and sector registry carry the same spectrum and sector distinctions.	Particle, resonance, confinement, or composite labels cannot float free of the certified state space.
Action-generator synchronization	1 of 18	Translation and Lorentz action identifiers match the corresponding self-adjoint generators and exponentiation maps.	The translation/Lorentz swap is rejected here.
Generator-Noether synchronization	1 of 18	The local current packet refers to the same generator system whose actions it is claimed to generate.	A current packet imported from an unrelated action cannot occupy the Noether role.
Noether-charge boundary product	1 of 18	Current, charge, integrability, whole-space falloff, and zero boundary flux hold jointly.	A local conserved current cannot be promoted automatically to a global charge.
CPT formulation match	4 of 18	Premises, geometry, state space, field algebra, implementation, and theorem conclusion instantiate the same admitted CPT formulation.	Inequivalent CPT routes cannot be mixed inside one certificate.
CPT carrier and sector synchronization	4 of 18	The antiunitary action operates on the certified state and field carriers, and charged-sector conjugation matches the spectrum registry.	CPT cannot act on a different carrier or silently alter the declared sector structure.

The exact conjunction is

$$\begin{aligned}
J_D = & J_{\text{geo/pres}} \wedge J_{\text{pres/pass}} \wedge J_{\text{geo/action}} \wedge J_{\text{action/state}} \wedge J_{\text{action/field}} \wedge J_{\text{state/field}} \\
& \wedge J_{\text{state/sectors}} \wedge J_{\text{action/generators}} \wedge J_{\text{generators/Noether}} \wedge J_{\text{Noether/charge}} \\
& \wedge J_{\text{CPTprem/geo}} \wedge J_{\text{CPTprem/state}} \wedge J_{\text{CPTprem/field}} \wedge J_{\text{CPTprem/action}} \\
& \wedge J_{\text{CPTact/state}} \wedge J_{\text{CPTact/field}} \wedge J_{\text{CPTact/charged}} \wedge J_{\text{sectors/charged}}.
\end{aligned}$$

No coordinate defaults to acceptance. Every applicable bridge is invoked and every failure receives a named diagnosis.

16.3.1 Four decisive noncollapse conditions

Projective state action versus finite-dimensional field covariance. The projective state representation acts on physical rays in the positive-metric state space. Local field covariance transforms field arguments and

components in a finite-dimensional representation. They share one Poincaré cover and must intertwine correctly, but they are not interchangeable occupants.

Boundary product status. Global charge standing is conjunctive. A local current and Ward identity are insufficient by themselves: the selected current family must match the claimed charge, the charge must be integrable, whole-space falloff must hold, and boundary flux must vanish. Failure of any factor leaves the local Noether architecture intact while blocking global-charge occupation.

CPT formulation match. The premise packet and antiunitary implementation must refer to the same admitted theorem formulation and conclusion. Convention phases and normalizations may vary inside a witnessed presentation class; the physical state space, field algebra, theorem conclusion, and sector action may not.

Synchronization. Synchronization is identity agreement across the interfaces that jointly describe one realization: geometry, state representation, active actions, generator system, current family, charge, field algebra, spectrum, and CPT conclusion. Pointwise eligibility does not guarantee these cross-role identities.

16.4 Complete assignment and local normal form

The executed certificate produces 2 complete pointwise-eligible assignments. The canonical assignment places the connected translation and Lorentz actions in their corresponding roles and synchronizes both with one Poincaré generator system. The only competing complete assignment exchanges the translation and Lorentz action occupants. It passes isolated fact checks but fails action–generator synchronization because the selected action identifiers no longer match the corresponding generator interfaces. Compatibility therefore leaves 1 complete assignment and 1 lawful orbit.

Table 16.2: Human-readable final assignment.

Role	Unique occupant	Why this occupant is required
Geometric domain	Minkowski/Poincare domain anchor	Fixes the Minkowski/Poincare spacetime target, causal order, orientation, and spin cover.
Presentation system	Coordinate, frame, and spin-frame presentation atlas	Supplies the coordinate, frame, and spin-frame atlases used to display the same target.
Passive redescription	Passive coordinate and frame redescription	Reexpresses coordinates or frames while preserving event and field identity.
Translation action	Connected translation action	Implements physical spacetime translations on the positive-metric state space.
Lorentz action	Connected Lorentz action	Implements physical rotations and boosts on the same state representation.
Projective state realization	Positive-metric projective state representation	Carries the unitary or antiunitary projective action on physical rays.
Local field covariance	Local covariant field realization	Transforms local field components and arguments without being identified with the ray action.
Sector response	Asymptotic and nonasymptotic sector registry	Classifies massive, massless, resonant, confined, and composite sectors in the certified state space.
Poincare generator bridge	Self-adjoint Poincare generator system	Links the active translation and Lorentz actions to their self-adjoint infinitesimal generators.

continued

Role	Unique occupant	Why this occupant is required
Noether local architecture	Renormalized local Noether-current package	Supplies local currents, Ward identities, and admitted improvement freedom.
Global charge and boundary realization	Whole-space charge and boundary realization	Adds integrability, whole-space falloff, and vanishing boundary flux needed for global charges.
CPT theorem premise	Jost/Wightman CPT premise packet	Fixes the Jost/Wightman premise packet that licenses the CPT theorem.
CPT physical implementation	Antiunitary CPT implementation	Implements the antiunitary CPT action on the certified physical states and local fields.

Audit-grade assignment ledger

The content-addressed role and residue identifiers, complete assignment map, closure counts, compatibility failure record, and automorphism data are printed in the Technical Audit Supplement, Appendices N–O.

16.5 Full-certificate and live automorphism groups

The full certificate and live assignment architecture are distinct theorem objects:

$$\Gamma_{\text{cert}} = \text{Aut}(R_D, X_D, E_D, J_D, \mathcal{R}_D), \quad \Gamma_{\text{live}} = \text{Aut}(R_D, X_D^{\text{live}}, E_D|_{\text{live}}, J_D|_{\text{live}}).$$

The restriction homomorphism

$$\rho : \Gamma_{\text{cert}}^{\text{live-preserving}} \longrightarrow \Gamma_{\text{live}}$$

induces the action used by the assignment theorem. Non-live certificate symmetries are recorded without being mistaken for live assignment equivalences.

Supplementary automorphism audit

The full-certificate group, live group, restriction image, and invariant-ablation results are printed in the Technical Audit Supplement, Appendix O.

SYM2-T71 and T75 — Dual automorphism and stable local closure

The full-certificate group, live group, and restriction image are computed separately. For the fixed certificate,

$$|\Sigma_C(D^\star)/\text{im } \rho| = 1.$$

This is the unique lawful normal form of the fixed fully certified target; it is not a uniqueness theorem for all relativistic quantum field theories or every possible realization of Poincaré symmetry and CPT.

16.6 Conservative certificate continuation

A conservative extension carries a target-preserving retraction to the canonical certificate and changes no standing burden, physical facet, quotient distinction, compatibility condition, failure route, or lawful orbit. Registered presentation and derived-witness enrichments preserve the singleton. New target, burden, facet, bridge, failure, or quotient content yields `ClosureReopened`.

Supplementary continuation audit

The conservative-extension retraction and closure-reopening ledger is printed in the Technical Audit Supplement, Appendix Y.

Local Endpoint, Necessary Kernel Activation, and Apex Closure

17.1 Imported-proposition path completeness

Every load-bearing primitive fact and bridge has at least one exact, canonically fixed, target-matched imported-proposition path containing a source identifier, proposition statement, hypotheses, conclusion facts, locator, specialization, and limitation. The checker verifies path completeness, target matching, and source and certificate integrity; the truth of each imported continuum theorem remains an explicit premise.

Table 17.1: Imported-proposition path completeness.

Evidence-path property	Result
All load-bearing facts have paths	PASS
Necessary support edges	96
Alternate support edges	282
Continuum theorems reproved	NO - imported interfaces

SYM2-T72 — Imported-proposition path completeness

No load-bearing physical fact or bridge is licensed solely by bibliographic association. Every such fact has a declared target-matched proposition path. The finite checker audits those paths but does not reprove the imported AASC, representation-theoretic, Noether, or CPT theorem.

17.2 Local finite theorem package

SYM2-T65–T70 — Construction, semantic use, presentation, and finite-interface refinement

The target is typed leaf by leaf; the physical universe is generated by the authoritative role-blind constructor grammar; presentation equivalence is witness-generated; and every registered physical refinement is executed through typed finite-interface projections and recompositions. The resulting refined normal forms transport bijectively to the canonical orbit. This is an interface-certificate theorem, not a constructive continuum-QFT existence result.

SYM2-T71–T75 — Dual automorphisms and stable local normal form

The complete eligibility relation, all 2 complete pointwise-eligible preassignments, every compatibility bridge, the full and live automorphism groups, the restriction image, and the assignment orbits are computed. The local assignment quotient is singleton.

17.3 Downstream apex consequence application

Kernel activation is complete before the apex application is considered. The fixed physical construction $D_{\text{SYM2}}^{\text{phys}}$ is non-degenerate and determinate, so the kernel and standing gate are instantiated by necessity; no downstream AMetric, apex, evidence, or software condition activates them. The finite certificate $C_{\text{SYM2}}^{\text{cert}}$ supplies the target-specific local occupation theorem and confirms that presentation changes, registered refinements, and conservative continuations preserve the same

target, while target-changing or standing-bearing additions reopen closure. The imported step is governed by the downstream proposition-level consequence interface in Table 1.2; apex status is unavailable if any required dependency identity, source, locator, or conclusion is absent.

The typed apex envelope is identified at the physical level:

$$D = (E, A, F, C, DAC).$$

- E.* Determinate physical states, fields, actions, carriers, sectors, records, and endpoints of the fixed SYM-2 target.
- A.* Kernel-licensed identity-preserving transformations, presentations, refinements, and continuations.
- F.* Fixed target, carrier, locus, and theorem identity throughout same-domain use.
- C.* The necessarily instantiated standing gate of the fixed non-degenerate determinate regime.
- DAC.* The forced same-domain closure of that standing-bearing regime under the inherited kernel, fixed-domain, and AMetric corpus machinery.

These are not manufactured by enumerating every software record or every finite composition of implementation functions. The certificate witnesses that the physical target remains the same domain and that the local normal form is nonempty, determinate, and unique. The upstream corpus supplies kernel necessity and the apex closure consequences.

Table 17.2: Necessary kernel activation and apex unique-interior consequence.

Closure property	Result
Non-degenerate determinate physical regime	PASS
Kernel instantiated by necessity	PASS
Same-domain application interface	PASS
Finite certificate faithfulness witnesses	PASS
Unique admissible interior	PASS
No second same-domain classifier	PASS
Scope	Kernel necessity follows from non-degenerate determinacy; the finite certificate fixes the local normal form; Unus Solus Possibilis Est supplies unique-interior closure.

SYM2-T76–T77 — Downstream apex consequence interface and unique admissible interior

The fixed local physical SYM-2 construction is non-degenerate and determinate and therefore necessarily kernel-bearing before this theorem is invoked. The exact fixed-domain, upstream AMetric-boundary necessity, apex-layer boundary compression, irreversible-commitment, composition, scope, meta-invariance, bivalence, target-fixation, and terminal uniqueness propositions are fixed by the downstream apex consequence interface; none is an additional kernel-activation condition. The standing-positive side is instantiated, the target is fixed, and the finite role-occupancy certificate has one lawful assignment orbit. Gate uniqueness, no-repair, no-generator, no-independent-authorizer, no illicit standing transfer, and target-fixation results therefore apply. Exactly one admissible interior exists; no second same-domain standing classifier, hidden selector, retrospective repair, or discrete plural admissible regime remains.

17.4 Capstone theorem

SYM2-T78 — Integrated Poincaré–CPT normal-form and apex closure

For SYM-2, the local physical construction is non-degenerate and determinate and therefore necessarily instantiates Admissibility, Standing, Reference, and Irreversibility. Authoritative physical-constructor saturation, typed semantic target use, isolated burden and candidate formation, finite-interface projection and recomposition, witnessed presentation quotienting, imported-proposition path completeness, complete incidence and relational compatibility,

dual automorphism closure, refinement stability, conservative-continuation discipline, and independent relational reconstruction determine one lawful local normal form. The imported apex theorem then forces one admissible interior and excludes every second same-domain standing classifier.

Verification statement

All 18 computed completeness clauses pass. The audit records 215 executed destructive mutations, 15 guard and dependency checks, and 9 aggregate reconstruction checks; every recorded check passes. The primary and shadow engines agree on the independently reconstructed finite theorem objects. Lean source is supplied only as a formal export path and is not counted as evidentiary support without a compiler log.

Worked Stress Tests and Dependency-Sensitive Falsification

18.1 Physical stress cases

The controlled cases include active translation versus passive origin shift; active boost versus inertial-frame redescription; projective phase versus physical state action; Hilbert-space representation versus finite-dimensional field covariance; stable particle versus resonance and confinement; local current versus global charge; CPT theorem versus implementation and convention phase; and curved-spacetime or alternative-theorem target updates.

18.2 Dependency-sensitive audit suite

The audit changes authoritative data and reruns the relevant pipeline. It includes constructor-grammar deletion and constructor omission; invalid target values and dead semantic consumers; unregistered or false bridge functions; broken presentation inverses and compositions; corrupted subgroup projections and parent recompositions; missing facets; target-fixation drift; unsupported facts; incomplete CPT premises; nonintegrable charges; injected second branches; full/live automorphism inconsistencies; collapse of the local gate to an empty or universal classification; attempted same-act repair; and target updates passed off as same-domain continuation.

Table 18.1: Dependency-sensitive audit classes.

Check class	Registered	Passed	Result
Destructive mutation	215	215	PASS
Guard/dependency	15	15	PASS
Aggregate reconstruction	9	9	PASS

Dependency-sensitive audit theorem

All 215 executed destructive mutations, 15 guard and dependency checks, and 9 aggregate reconstruction checks pass. A destructive-mutation row changes authoritative input and requires the relevant calculation to detect the change. Guard rows verify fixed dependency or scope conditions without being mislabeled as mutations. Aggregate rows recompute complete theorem objects or architecture families. The suite establishes sensitivity to the registered failure classes; it does not replace the open falsification route for a genuinely new same-target burden, constructor, physical distinction, or kernel-breaking same-domain counterexample.

Regression success is not an imported physics proof

The mutation suite tests the finite certificate and its application interface. It does not prove the external continuum propositions or generate the AASC kernel from software output.

Audit Architecture, Formal Boundary, and Reproducibility

19.1 Controlled research objects

The project includes editable LaTeX sources; strict target and certificate data; isolated role and physical-object builders; the authoritative constructor grammar; parent-object, facet, witness, proposition, eligibility, compatibility, assignment, automorphism, refinement, continuation, kernel-activation, and apex-interface records; generated JSON, CSV, and LaTeX projections; mutation, guard, and reconstruction checks; source manifests; and formal specification files. The manuscript is a rendered projection of those controlled objects.

19.2 Build order

1. Validate and type the target; certify local non-degenerate determinacy; apply necessary kernel activation.
2. Derive burdens and canonical roles in the isolated role process.
3. Construct physical parents and iterate the authoritative constructor grammar to a fixed point in the isolated physical process.
4. Execute presentation witnesses and imported-proposition paths; form the witnessed presentation residue; classify every role-residue pair.
5. Invoke every bridge; compute compatibility, the live residue, full and live automorphisms, complete assignments, and assignment orbits.
6. Rebuild every registered finite-interface refinement and conservative continuation.
7. Certify the same-domain apex application interface and render the publication objects.

No downstream assignment, role label, or eligibility result is reloaded as an upstream physical premise.

19.3 Dependency-sensitive testing

The audit separates executed destructive mutations from guard and dependency checks and from aggregate reconstruction checks. It mutates every target-defining leaf, required constructor and parent declaration, every bridge consumer, every presentation witness, every registered projection/recomposition family, and every proposition-exact apex provenance field. The provenance mutations replace statements, hypotheses, conclusions, specializations, limitations, prerequisite edges, source titles, DOI metadata, and bundled-file hashes; they also introduce missing and cyclic prerequisites and false local-verification claims. Guard rows verify process isolation, kernel activation, and same-domain scope without being represented as destructive mutations. Aggregate rows recompute automorphism structures, refinement families, conservative continuations, and the independent shadow certificate.

The class totals are reported in Table 18.1. The complete row-level ledger is supplied in the technical audit supplement and machine-readable project record.

19.4 Independent relational reconstruction

The shadow checker begins from primitive target, burden, physical, and proposition inputs rather than the primary engine's role, residue, eligibility, or assignment outputs. It independently reconstructs governance classes, physical facets, positive incidence,

cross-interface compatibility, and the singleton assignment using relational tables and independently coded predicates. Its claim is limited to the theorem objects it actually reconstructs.

Table 19.1: Independent primitive-input relational checker.

Independent-checker property	Result
role_count	13
facet_count	13
eligibility_pair_count	169
preassignment_count	2
compatible_assignment_count	1
role_keys_match	PASS
assignment_role_keys_match	PASS
singleton_orbit	PASS
intermediate_digests	3/3 PASS
all_pass	PASS
scope	Independent upstream relational reconstruction from target and burden primitives; presentation variants and full graph automorphisms remain primary- engine audits.

19.5 Formal boundary

The two executed Python engines provide the finite certificate. Lean source is supplied as a formal export/specification route and is not cited as evidence without a compiler log. Continuum AASC, representation-theoretic, Noether, and CPT results remain imported theorem interfaces.

Exact proof boundary

The theorem domain is the fixed physical SYM-2 regime; the project is its finite faithful certificate. The finite proof establishes local closure relative to the frozen typed target and registered burden, physical-constructor, presentation, refinement, rival, and continuation grammars. It witnesses non-degenerate determinacy and the same-domain application interface. It does not manufacture the kernel, AMetric boundary, apex theorem, or imported physical propositions, and it does not claim open-world completeness beyond the controlled grammars.

19.6 Publication objects

The project contains independently compilable journal article, technical audit supplement, and combined audit manuscript. The complete archive contains editable sources and the executable certificate. The Overleaf archive contains editable manuscript sources and generated publication tables rather than a wrapper around a precompiled PDF.

Conclusion

Spacetime symmetry is not one object. Minkowski/Poincaré structure fixes the flat target. Active translations, rotations, and boosts act physically. Passive coordinate, frame, spin-frame, ray-phase, current-improvement, and CPT-convention changes are skin only through validated witnesses. Projective state representation, local field covariance, stable particle response, resonance, confinement, generator, current, charge, Ward identity, CPT premise, theorem implication, physical implementation, charged-sector consequence, and convention phase occupy distinct loci.

AASC supplies the necessity architecture under which those distinctions can be determinate at all. The local physical construction is non-degenerate and determinate, so Admissibility, Standing, Reference, and Irreversibility are instantiated by necessity rather than added as an interpretive layer. Conventional geometry and quantum field theory supply target-specific propositions within that kernel-bearing domain. Licensed representation does not imply unreality: quotient-stable geometry, states, actions, representation classes, generators, charges, and CPT structures are real target-internal content.

The local finite endpoint is exact. 27 target-derived burdens yield 13 canonical roles. An isolated role-blind process closes the authoritative physical-constructor grammar over 15 parent objects. 39 generated records form 30 witnessed presentation residues. All 390 role-residue pairs are classified; 15 positive edges generate 2 complete pointwise-eligible preassignments; relational compatibility retains 1 complete assignment. The full-certificate group, live group, and restriction image have orders 1, 1, and 1 respectively, and the induced assignment quotient has 1 orbit. Every one of the 64 registered refinement architectures is rebuilt through typed finite-interface projection and recomposition and transports to the canonical normal form.

The canonical and normalized Jost/Wightman CPT presentations induce the same normal form; unbridged algebraic and Lagrangian formulations remain distinct certificate targets. The certificate is a faithful finite witness of the fixed physical domain, not a separate bookkeeping domain. Kernel activation is complete because that physical regime is non-degenerate and determinate. The subsequently fixed-domain, composition, scope, meta-invariance, AMetric-boundary, bivalence, target-fixation, and terminal uniqueness propositions are downstream consequence interfaces, not activation conditions. With those interfaces and the local singleton normal form fixed, the unique-admissible-interior theorem of *Unus Solus Possibilis Est* applies. No second same-domain classifier, hidden selector, retrospective repair, or discrete plural admissible interior remains.

Capstone statement

Relative to the fixed fully certified target SYM-2, the kernel-faithful, constructor-saturated, semantically causal, projection-recompositional SYM-2 construction has one lawful local normal form; completely types every object and facet generated by the registered physical grammar; is invariant under the registered presentation, finite-interface refinement, and conservative-continuation systems; preserves CPT normal form at exactly the proved relation strength; and, after complete necessary kernel activation, by the downstream imported apex consequence chain lies in one unique admissible interior with no residual same-domain standing classifier.

Supplementary Audit Contents

The journal article contains the complete proof spine, theorem statements, local normal-form result, dependency order, and exact scope. Full registries and row-level audit objects are supplied in the separately compilable *Technical Audit Supplement* and the machine-readable project archive.

Supplement location	Complete record
Appendices A–J	Canonical facet atlas; integrated target registry; theorem and source inventories; burden certificates; active/passive, representation, generator, Noether, and CPT locus certificates.
Appendices K–O	Target-leaf, bridge, constructor, projection, presentation, eligibility, compatibility, assignment, automorphism, and orbit ledgers.
Appendices P–T	Formal boundary, cost of kernel denial, mutation and reconstruction records, exact nonclaims, and project architecture.
Appendices U–Z	CPT formulation covariance, no-second-classifier route, computed completeness, refinement stability, certificate continuation, evidence paths, shadow reconstruction, and final apex closure.
Project archive	Authoritative JSON and CSV records, executable primary and shadow engines, proposition-exact provenance manifest, source-integrity statuses, formal export, build logs, and checksums.

The supplement is evidentiary documentation for the same fixed physical theorem domain; it is not a second theorem domain or an additional condition for kernel activation.

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